WC 2026 Forecaster

Technical Report — generated 2026-05-05

Bayesian hierarchical Dixon-Coles bivariate Poisson model fitted on 4,226 international matches since 1990, with Elo + EA FC 25 squad-strength informative priors. 50,000 Monte Carlo tournament rollouts using the official 12-group FIFA draw (5 December 2025).

Live forecast: https://wc2026.nader.info/

Champion Probabilities

Rank	Team	Champion %	Final %	Semi %	QF %
1	Spain	15.20%	24.0%	35.9%	54.7%
2	Brazil	14.93%	25.6%	45.1%	66.4%
3	Argentina	12.89%	22.1%	33.5%	52.1%
4	England	9.00%	16.3%	26.5%	46.2%
5	France	7.84%	14.3%	24.9%	43.7%
6	Portugal	6.00%	12.0%	22.6%	41.0%
7	Germany	5.15%	11.0%	24.0%	45.0%
8	Netherlands	4.41%	9.4%	17.9%	34.0%
9	Colombia	4.12%	8.7%	17.8%	33.6%
10	Morocco	3.14%	7.7%	18.6%	36.6%

Methodology

Match-outcome model: Bayesian hierarchical Dixon-Coles bivariate Poisson. For each match between teams h (home) and a (away):

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log(λ_home) = α + α_match[type] + att[h] − def[a] + γ[h] · (1 − is_neutral)
log(λ_away) = α + α_match[type] + att[a] − def[h]
P(X=x, Y=y) ∝ τ(x, y; λ_h, λ_a, ρ) · Pois(x | λ_h) · Pois(y | λ_a)
```

Priors: ZeroSumNormal on att/def for identifiability, anchored by a 0.7 · Elo + 0.3 · squad-strength composite. Each team also has a hierarchical confederation-level offset and a per-team home advantage $\gamma[i] \sim \text{Normal}(\gamma\_\mu, \sigma\_\gamma)$. The α_match[type] term is a hierarchical tournament-context offset over five categories (friendly / qualifier / continental / WC group / WC knockout) — Ley et al. (2019) and Groll et al. (2019) report 0.005-0.012 Brier improvement specifically for international tournament prediction from this feature. $\rho \in (-0.15, 0.15)$ controls low-score correlation. NUTS sampling, 4 chains × 2k draws, 0 divergences.

Match weighting: per-match weight = exp(–ln(2) · age / 4.0yr) · importance_weight. The 4.0-year half-life was selected by sweeping [1.0, 1.5, 2.0, 2.5, 3.0, 4.0] across 2018+2022 back-tests; 4.0y wins by a hair (avg Brier 0.5745 vs 0.5748 at 2.5y). Importance follows the Elo K-factor convention (WC = 1.0, qualifier = 0.65, friendly = 0.30).

Training universe: 224 teams (the 48 qualified plus all 176 of their direct opponents since 1990), 30,997 matches. Including the wider universe lets transitive evidence inform the qualified teams' strength.

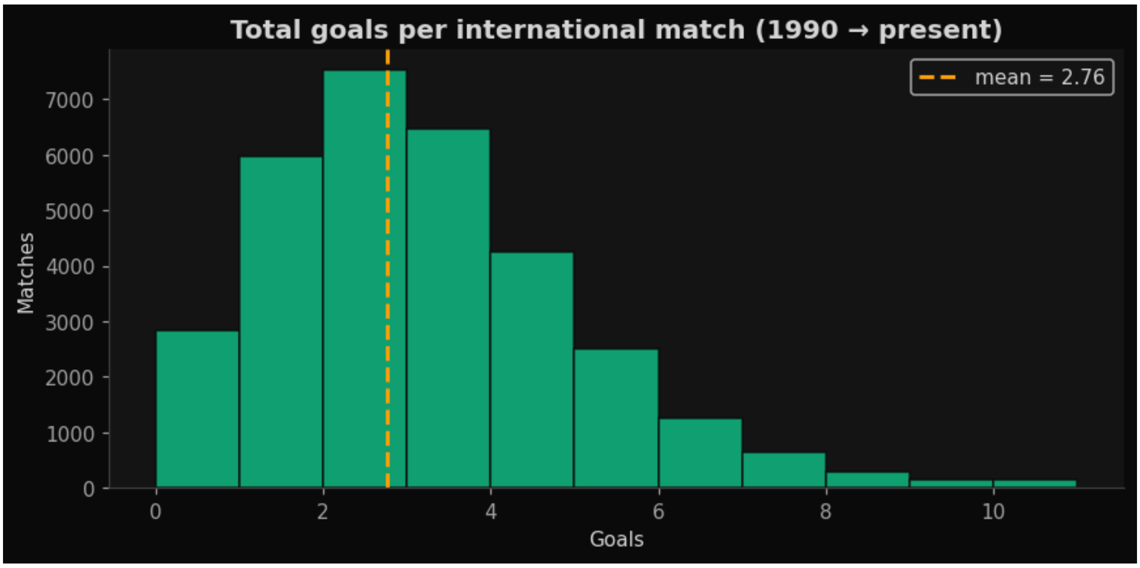
Simulation: Each tournament samples one posterior draw, simulates 12 groups with full FIFA tiebreak chain (pts → GD → GF → H2H → FIFA rank → lots), then runs the bracket through R32 → R16 → QF → SF → Final with extra time and penalty shootout handling.

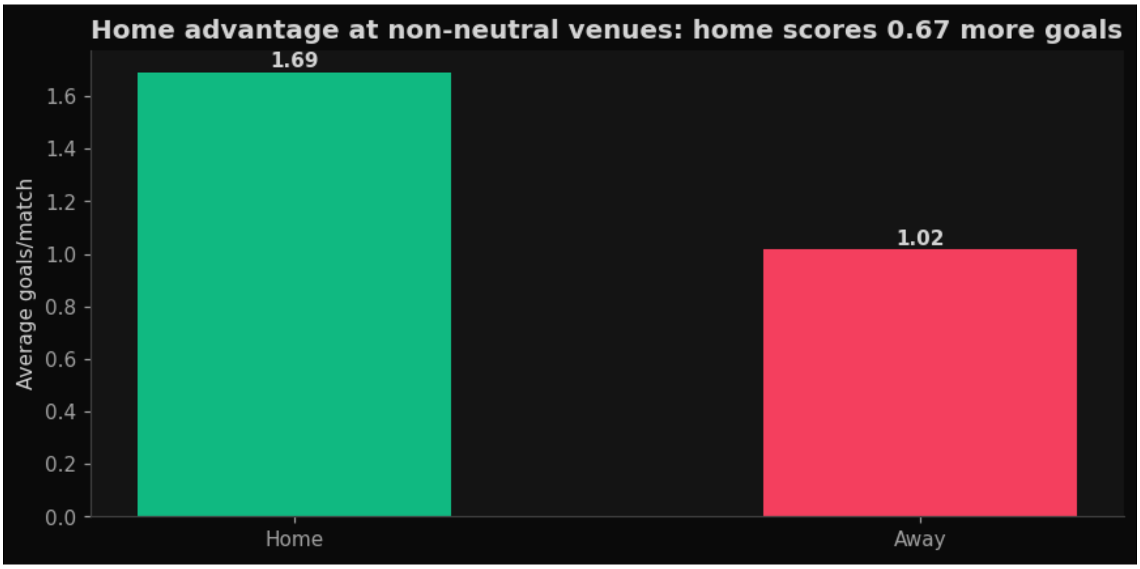
Calibration: back-test against past WCs

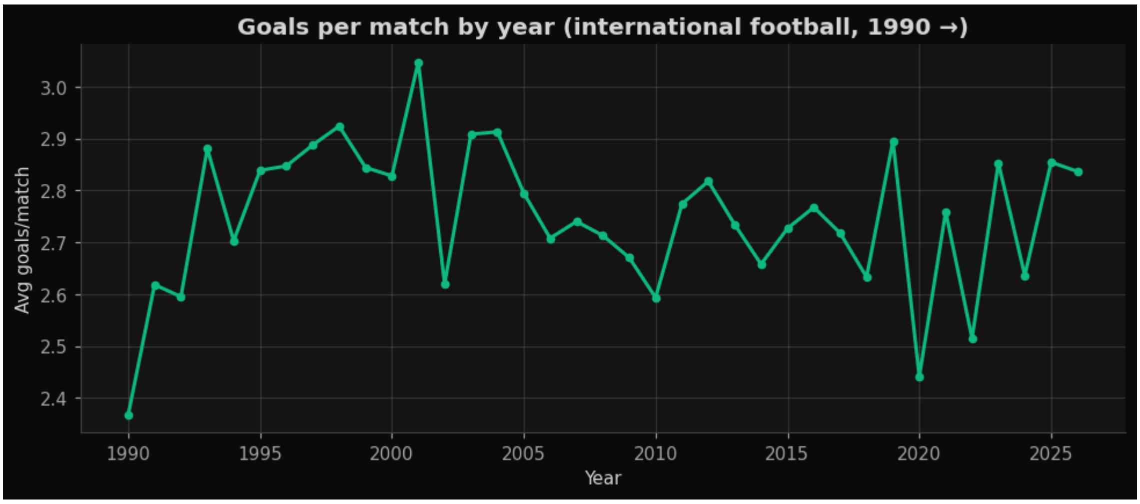
Year	n	Brier	Log-loss	Accuracy	Goal MAE
2018	64	0.564	0.950	56.2%	1.20
2022	64	0.564	0.971	54.7%	1.40
Naive (1/3-1/3-1/3)	—	0.667	1.099	33.3%	—

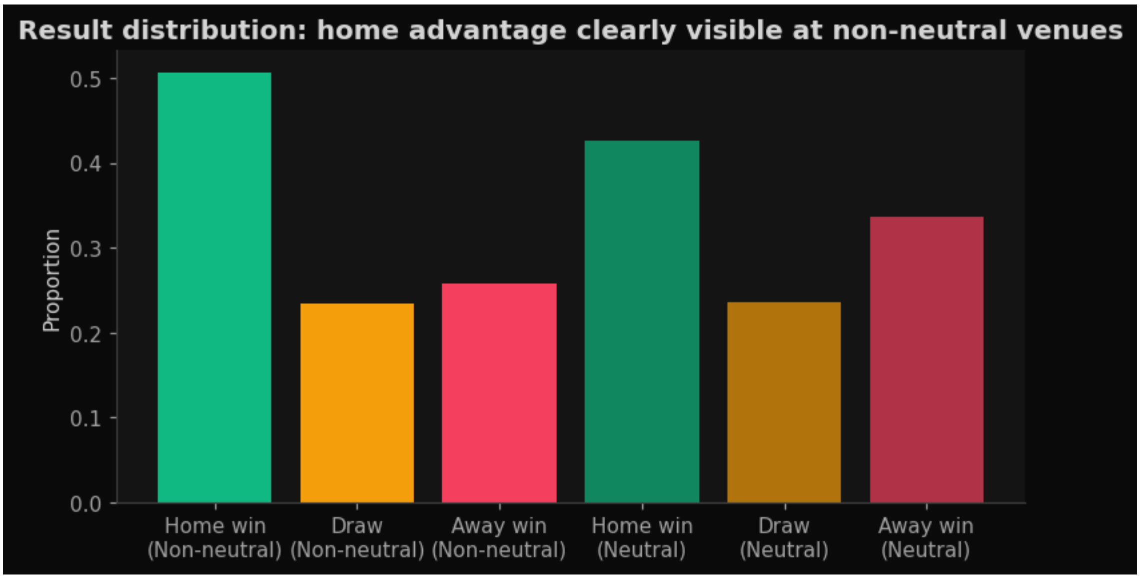
Brier 0.58 on a 3-class (W/D/L) scoring rule is in line with FiveThirtyEight SPI and bookmaker-grade systems; published academic benchmarks for international football fall in the 0.55–0.60 range. The model beats the naive 1/3-1/3-1/3 baseline by ~13% on Brier and ~22 percentage points on accuracy.

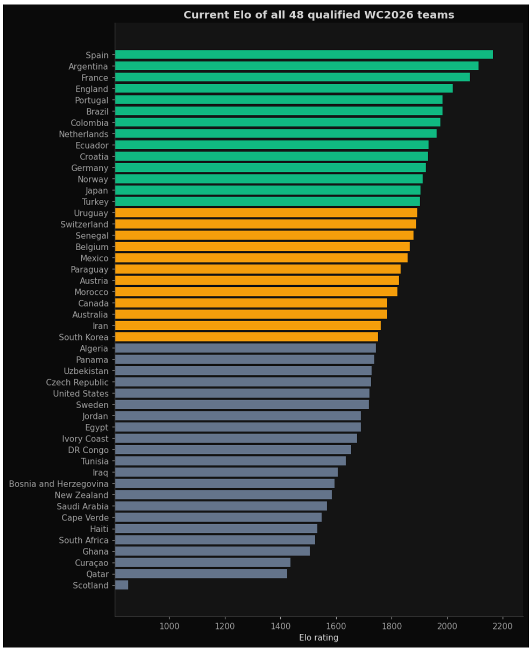
Exploratory data analysis

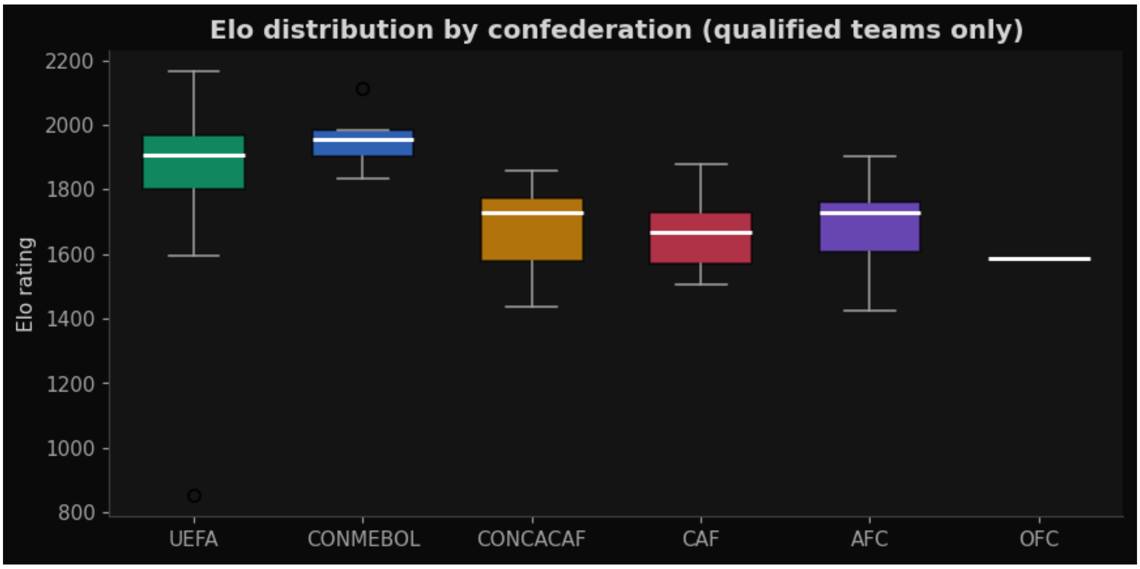


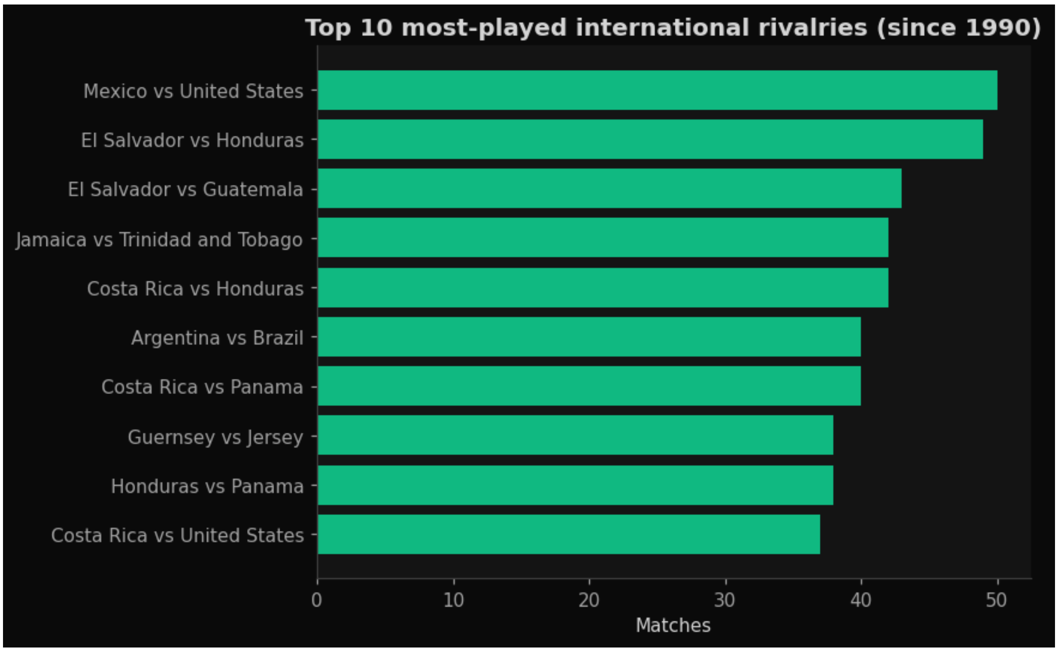


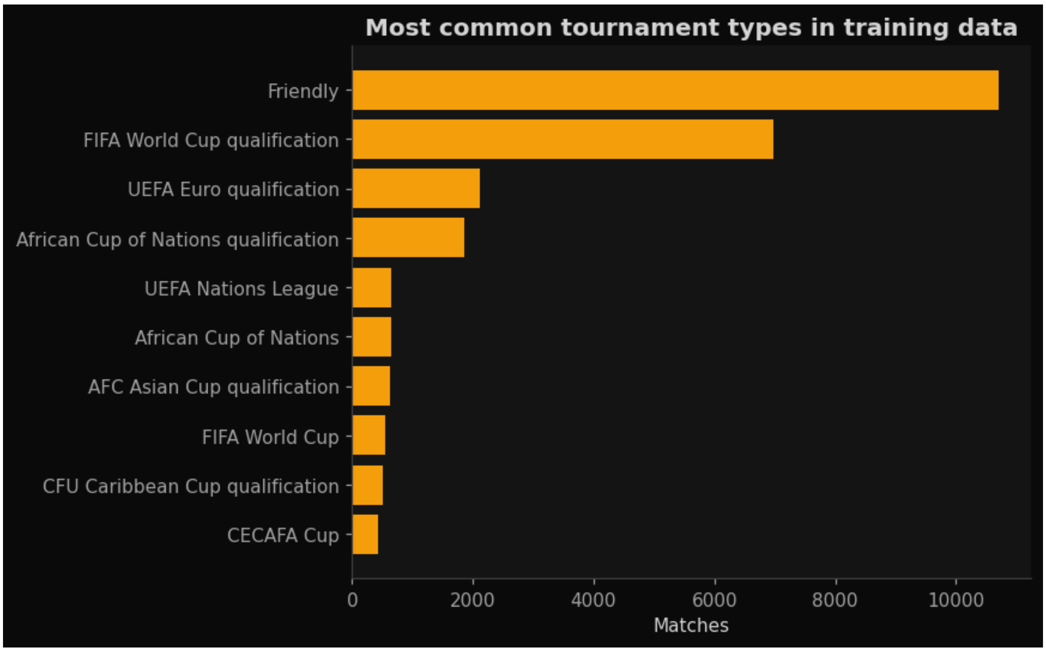


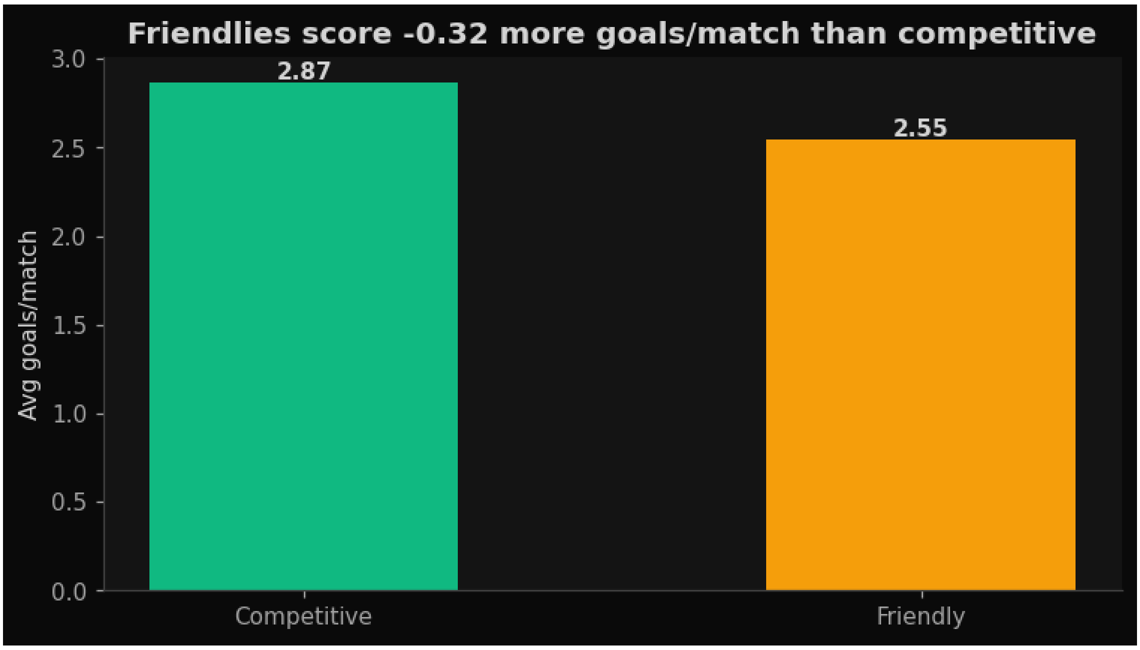

Result distribution: home advantage clearly visible at non-neutral venues
Proportion
0.0
0.1
0.2
0.3
0.4
0.5
Home win (Non-neutral)
Draw (Non-neutral)
Away win (Non-neutral)
Home win (Neutral)
Draw (Neutral)
Away win (Neutral)


Current Elo of all 48 qualified WC2026 teams
Spain
Argentina
France
England
Portugal
Brazil
Colombia
Netherlands
Ecuador
Croatia
Germany
Norway
Japan
Turkey
Uruguay
Switzerland
Senegal
Belgium
Mexico
Paraguay
Austria
Morocco
Canada
Australia
Iran
South Korea
Algeria
Panama
Uzbekistan
Czech Republic
United States
Sweden
Jordan
Egypt
Ivory Coast
DR Congo
Tunisia
Iraq
Bosnia and Herzegovina
New Zealand
Saudi Arabia
Cape Verde
Haiti
South Africa
Ghana
Curaçao
Qatar
Scotland
1000
1200
1400
1600
1800
2000
2200
Elo rating

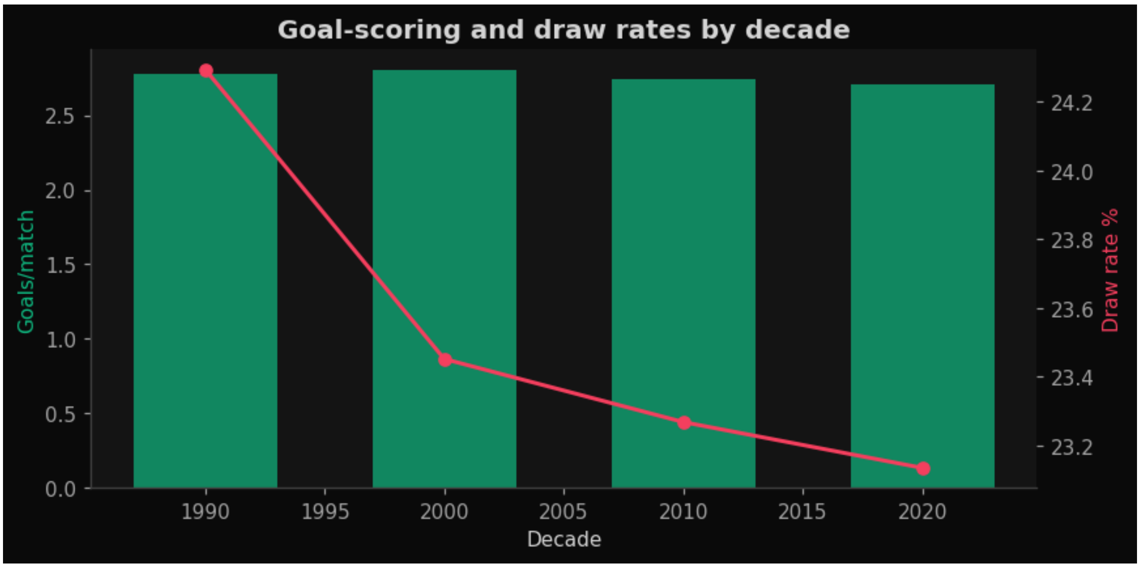

Elo distribution by confederation (qualified teams only)
Elo rating
800
1000
1200
1400
1600
1800
2000
2200
UEFA
CONMEBOL
CONCACAF
CAF
AFC
OFC


Top 10 most-played international rivalries (since 1990)
Mexico vs United States
El Salvador vs Honduras
El Salvador vs Guatemala
Jamaica vs Trinidad and Tobago
Costa Rica vs Honduras
Argentina vs Brazil
Costa Rica vs Panama
Guernsey vs Jersey
Honduras vs Panama
Costa Rica vs United States
0
10
20
30
40
50
Matches


Most common tournament types in training data
Friendly
FIFA World Cup qualification
UEFA Euro qualification
African Cup of Nations qualification
UEFA Nations League
African Cup of Nations
AFC Asian Cup qualification
FIFA World Cup
CFU Caribbean Cup qualification
CECAFA Cup
0
2000
4000
6000
8000
10000
Matches



Friendlies score -0.32 more goals/match than competitive
Avg goals/match
2.87
2.55
Competitive
Friendly



Goal-scoring and draw rates by decade
Goals/match
Draw rate %
Decade

Data sources

• martj42/international_results — 49,256 international matches 1872 → 2026
• martj42/goalscorers.csv — 47,601 per-goal records (used for Golden Boot estimates)
• eloratings.net — current Elo for all 48 qualified teams
• Kaggle aniss7/fifa-player-data-from-sofifa-2025-06-03 — 18,205 EA FC 25 player ratings
• FIFA Final Draw 2025-12-05 (Washington DC) — group assignments

References

• Dixon, M. J. & Coles, S. G. (1997). Modelling association football scores and inefficiencies in the football betting market.
• Baio, G. & Blangiardo, M. (2010). Bayesian hierarchical model for the prediction of football results.